2019 EXAMINATIONS

Lancaster University

Part II

COMPUTING AND COMMUNICATIONS [2.5 Hours]

SCC.341 Game Design

Candidates are asked to answer THREE questions from FOUR; each question is worth a total of 25 marks.

Use a separate answer book for each question.

Question 1

Assume that you are consulting for Niantic. They want to explore options for communication between players for their flagship game Pokemon GO.

1.a Identify the different in-game communication options that are feasible and explain why these are feasible

[5 marks]

1.b Discuss the design considerations that influence the choice of in-game communication used.

[5 marks]

1.c Recommend a specific option and contrast its use with existing mechanisms like Discord and Telegram.

[5 marks]

1.d Consider the block diagram of a communication systems as shown in Figure 1.d below:

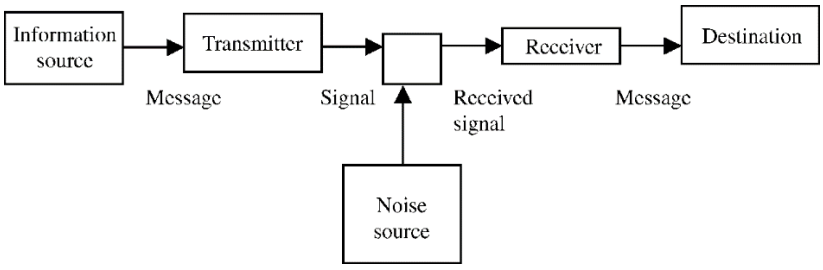

Figure 1.d: Block diagram of a communication system

i. How does information flow from component to component in Figure 1.d above?

[3 marks]

ii. With the aid of examples, explain how the components of the communication system, as shown in Figure 1.d, are manifested in a game.

[3 marks]

1.e Consider the use of information theory in games:

i. How are uncertainty (or degree of) and information theory linked?

[2 marks]

ii. How does information theory as a concept connect with games?

[2 marks]

Total 25 marks

Question 2

2.a What is a zero-player game? Identify two examples of zero player games.

[5 marks]

2.b How would you use the concepts of a zero-player game to design the flow channel of a game?

[5 marks]

2.c Consider a simple cybernetic system:

i. Explain through an example the ways in which such a system gauges its effects and makes necessary adjustments.

[3 marks]

ii. Design a function-based model based on a cybernetic system for a game. With the aid of an example, explain how the functions of each state are linked.

[6 marks]

2.d From a design viewpoint, games should consider feedback mechanisms.

i. What are the design rules relating to feedback in games?

[3 marks]

ii. Explain using relevant examples the effect these feedback mechanisms have on a game's stability, balance and outcome.

[3 marks]

Total 25 marks

Question 3

3.a Shigeru Miyamoto identifies five important game design principles:

- Start with a simple concept
- Design around the computer's limitations
- Minimize the player's confusion
- Incorporate a smooth learning curve
- Accommodate all skill levels

Jesse Schell talks about game design using the concept of a tetrad consisting of:

- Mechanics
- Aesthetics
- Story
- Technology

Using specific examples of games (and specific elements within that specific game), demonstrate how each one of Miyamoto's principles relate to elements within Schell's tetrad. (3 marks per principle)

[15 marks]

3.b The relationship between a game decision and a game outcome can have different levels of uncertainty. Identify each of the levels of uncertainty and explain how they are manifested using relevant examples.

[3 marks]

3.c Consider a dice based game where its goal and rules are detailed as follows:

- Goal: to avoid rolling 6s and to be the first player who reaches 50 points.
- Rules:
  - If you roll a 6, you lose your turn and do not score.
  - If you roll any other number, you receive the corresponding points. You may throw as often as you wish and accumulate the points. If, however, you roll a 6, all your accumulated points will be lost.

i. What does this game offer to a player? **[1 mark]**

ii. How is balance used in this game? **[1 mark]**

Question 3 continues on next page…

Question 3 continued.

iii. What is the probability of rolling a 6?

a. When a player decides to roll once then roll twice in succession? **[1 mark]**

b. What does this tell us about the likelihood of rolling a 6 and what can a player deduce from this? **[1 mark]**

c. What does this tell us about the player's decision to roll or not to roll?

[1 marks]

iv. What would be the recommended strategy for a player to follow?

[2 marks]

Total 25 marks

Question 4

The screenshot below shows a scene from Angry Birds wherein the bird is catapulted towards the pigs' building. Your objective is to ensure that the aesthetics of asset behaviour is maintained. Use this information to answer Questions 4a and 4b.

4.a What is the key principle that guides your design?

[2 marks]

4.b Identify 4 critical elements related to game-physics that you (as a designer) would ask a developer to implement.

[8 marks]

4.c Using the definition of a system and its shared elements, explain the different ways a game can be framed as a system.

[6 marks]

4.d Based on a game example, discuss the design considerations that need to be addressed given the different ways the elements in the system are framed.

[6 marks]

4.e Discuss a method that can be used to link design, systems and meaningful play in understanding the challenges presented to game design.

[3 marks]

Total 25 marks

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